

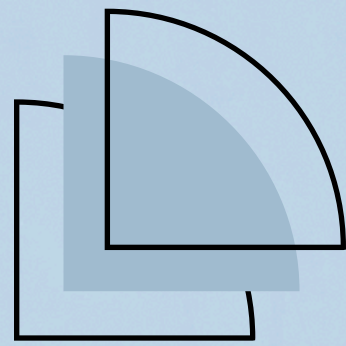
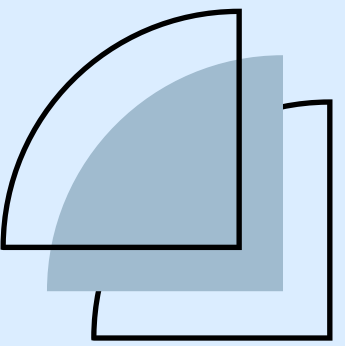
Reversible Data Perturbation for Privacy Preservation in IIoT Systems: Lightweight Reversible Perturbation for IIoT Telemetry

Group Members:

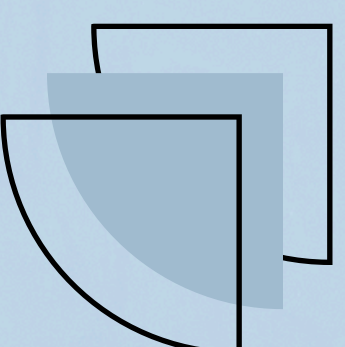
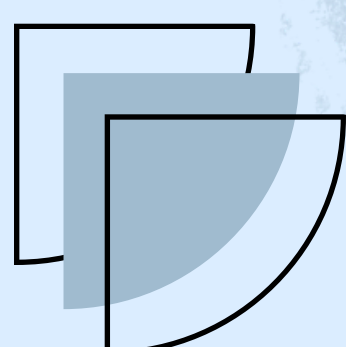
1. **RABBI HASAN HIMEL**
2. **TAYYABA BASRI**
3. **HAMZA HAROON**

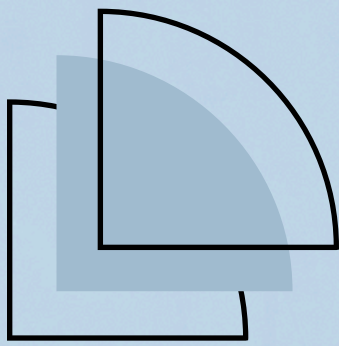
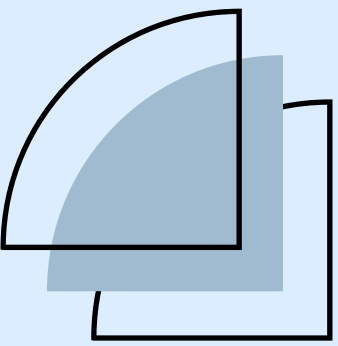
Supervised by:

Doç. Dr. Emrullah Fatih
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OBJECTIVES

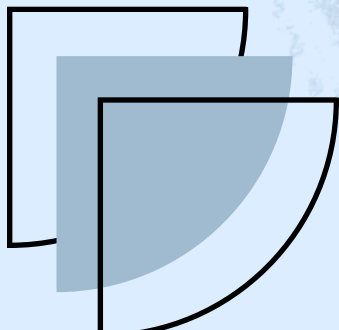
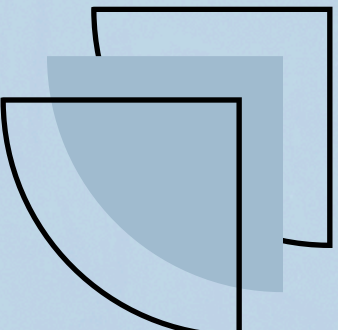
- Study existing data perturbation techniques.
 - Evaluate reversible perturbation methods for IIoT data.
 - Analyze the privacy–utility trade-off between traditional encryption, numerical noise addition, and bitwise approaches.
 - To identify the most efficient perturbation method that delivers strong privacy.
 - Compare bitwise (XOR) and numerical (Gaussian) perturbation for privacy
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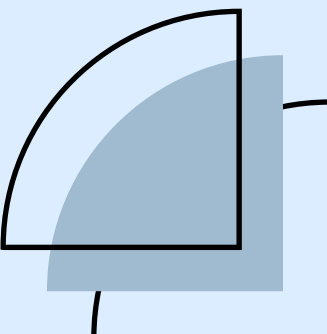


INTRODUCTION

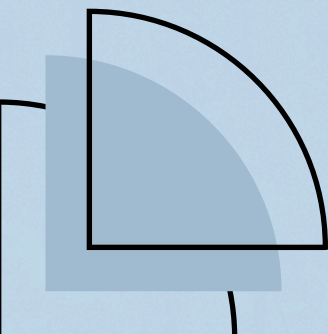
- IIoT systems continuously share sensor and machine data with cloud platforms
- Sharing raw industrial data creates serious privacy and security risks
- Traditional encryption methods are computationally heavy for legacy devices
- Numerical noise-based perturbation distorts data and reduces accuracy
- Industrial systems require a lightweight and reversible privacy solution

This work focuses on XOR-based reversible perturbation for 16-bit Modbus data





PROBLEM STATEMENT

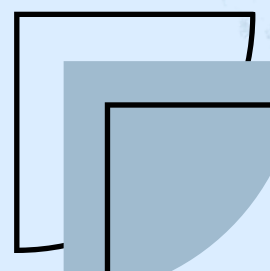
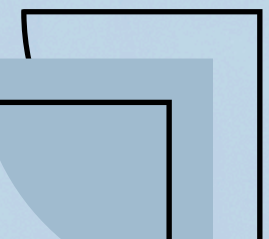


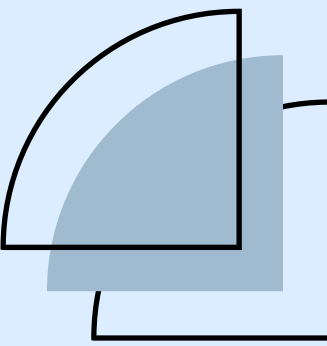
IIoT systems continuously stream real-time industrial telemetry to the cloud (e.g., Modbus 16-bit registers) for monitoring, analytics, and machine learning.

KEY PROBLEMS

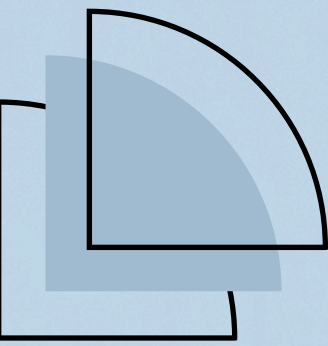
- Data exposure risk
- Legacy hardware limits: 16-bit controllers demand ultra-low latency and minimal CPU work.
- Gaussian noise does not provide sufficient privacy, still above random guessing
- AES / homomorphic encryption impose high CPU, latency, and energy cost.
- Many perturbation methods prevent exact recovery or distort values, degrading cloud analytics and ML.

Breaking control-loop timing or ML accuracy is not acceptable in industrial systems..





DESIGN REQUIREMENTS

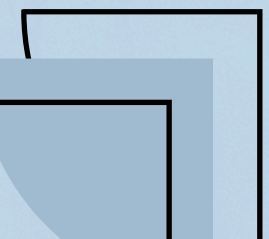


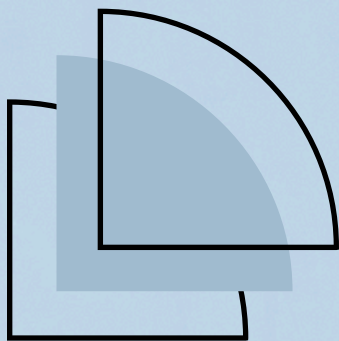
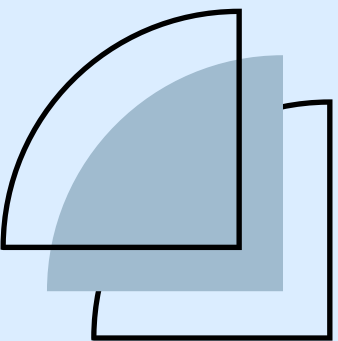
OPERATIONAL REQUIREMENTS

- Strong privacy protection
- Minimal latency
- Data utility
- Strong privacy with reversibility

SYSTEM CONSTRAINT

Existing encryption mechanisms are not designed for native 16-bit Modbus telemetry and introduce significant computational and energy overhead.





PROPOSED SOLUTION

PROPOSED SOLUTION

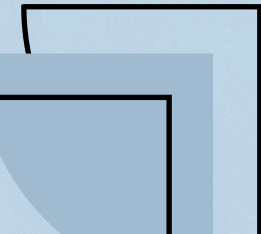
We introduce Lightweight Reversible Data Perturbation (RDP) for Industrial IoT streams, designed specifically for legacy 16-bit devices.

- XOR on native 16-bit registers
- Pseudo-Random Key Stream (PRNG)
- Perfect Reversibility

TECHNICAL ADVANTAGE

- One CPU Instruction
- Native 16-bit Compatibility
- Real-Time Safe

OUTCOME

- Secure data sharing without encryption overhead
 - 100% data recovery in the cloud
 - Preserves full analytics and ML accuracy
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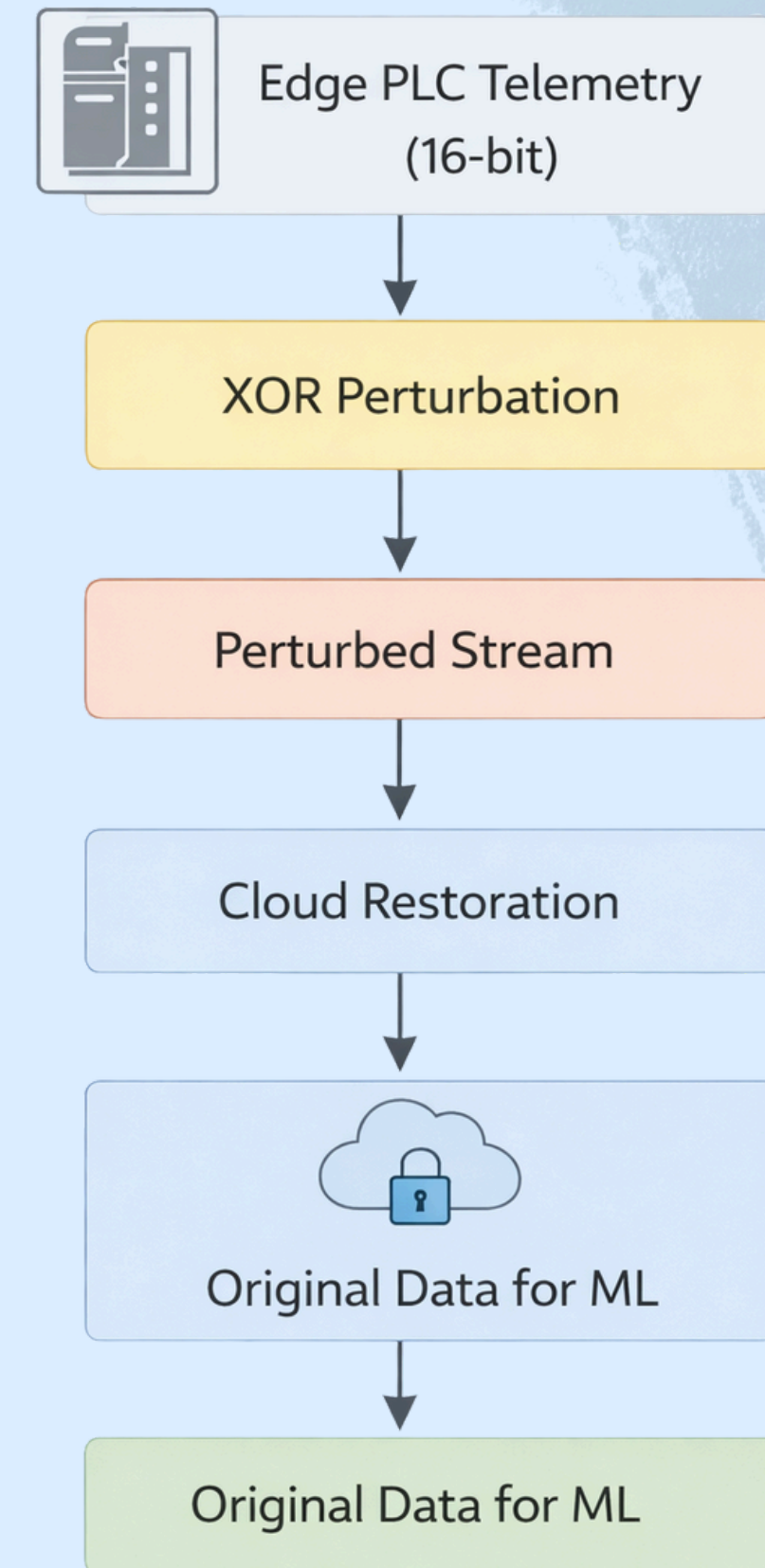
COMPARISON BETWEEN XOR AND NUMERICAL PERTURBATION

XOR	NUMERICAL PERTURBATION
• USES A BITWISE OPERATION INSTEAD OF CHANGING VALUES • DOES NOT ALTER THE DATA STRUCTURE • ALLOWS PERFECT RECOVERY OF ORIGINAL DATA • PRESERVES 100% DATA ACCURACY AFTER RESTORATION • VERY FAST AND LIGHTWEIGHT, SUITABLE FOR REAL-TIME IIoT • NUMERICAL PERTURBATION FAILS TO HIDE STATISTICAL PATTERNS	• ADDS RANDOM VALUES TO THE DATA • CHANGES THE ORIGINAL NUMERICAL VALUES • CAUSES LOSS OF ACCURACY • ORIGINAL DATA CANNOT BE PERFECTLY RECOVERED • NOT IDEAL FOR INDUSTRIAL SYSTEMS THAT NEED EXACT VALUES • WHILE XOR DESTROYS THEM.

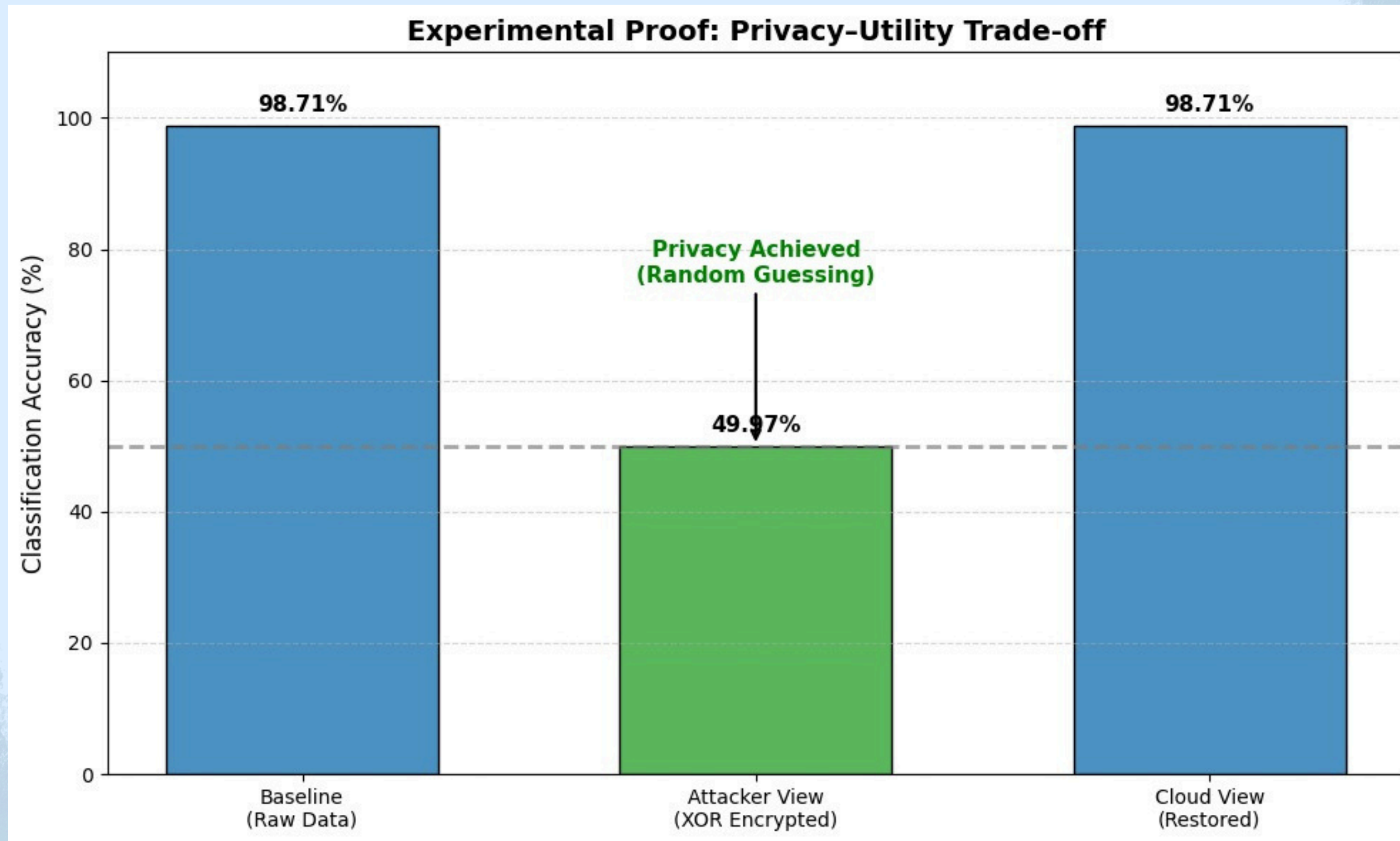
METHODOLOGY

(The XOR Framework) - How it works

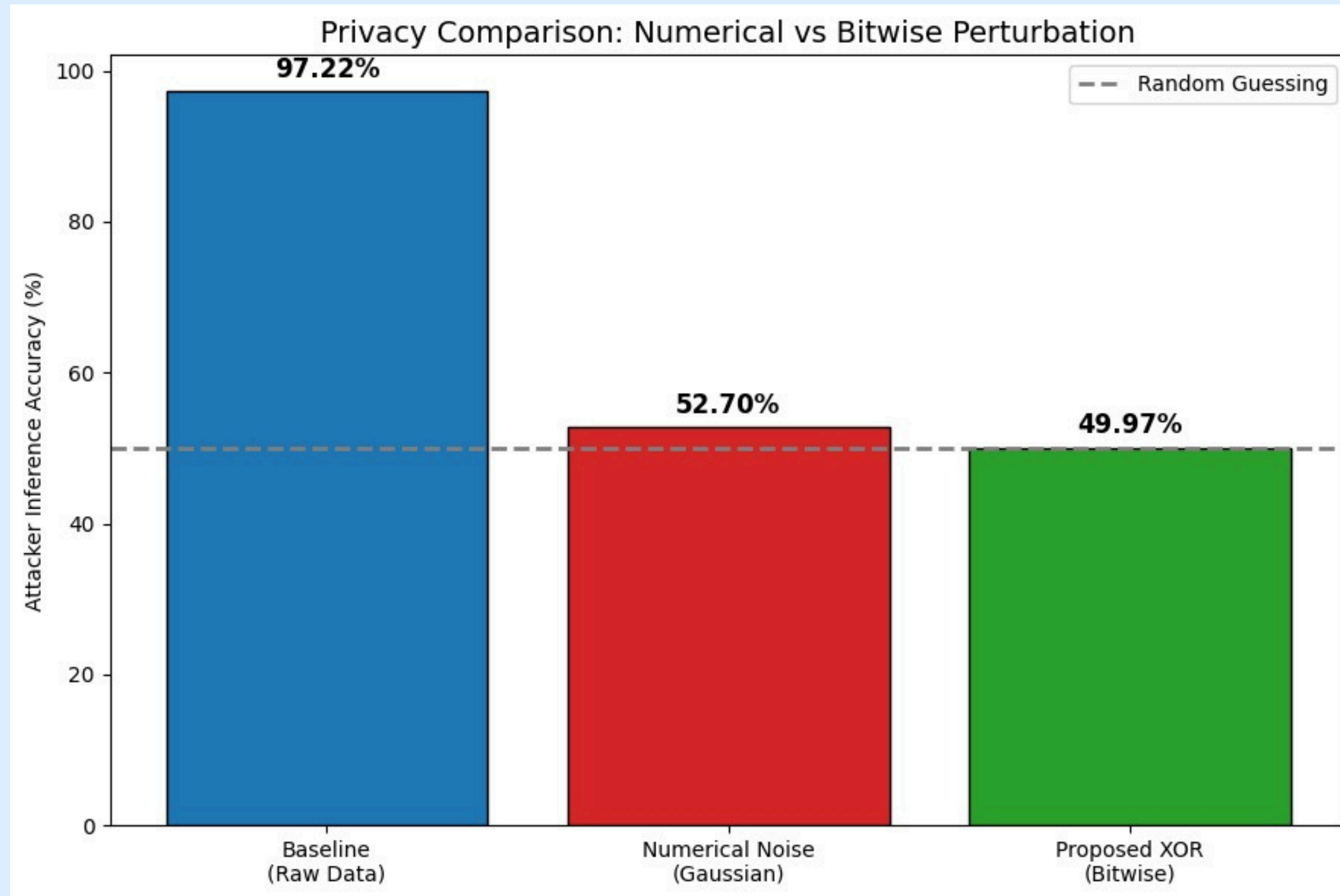
- Data Type: Native 16-bit Modbus Integers (0 - 65535).
- The Algorithm: **Bitwise XOR $\oplus$**
- Edge (Factory): **Raw Data $\oplus$ Key = Scrambled Noise**
- Cloud (Server): **Scrambled Noise $\oplus$ Key = Original Data**
- Why Bitwise XOR (Design Choice)
 - Hardware Authentic (Native to PLCs).
 - Zero Data Loss (No rounding errors).
 - Single-cycle CPU operation (ultra-low latency).



PRIVACY-UTILITY TRADE-OFF (EXPERIMENTAL PROOF)

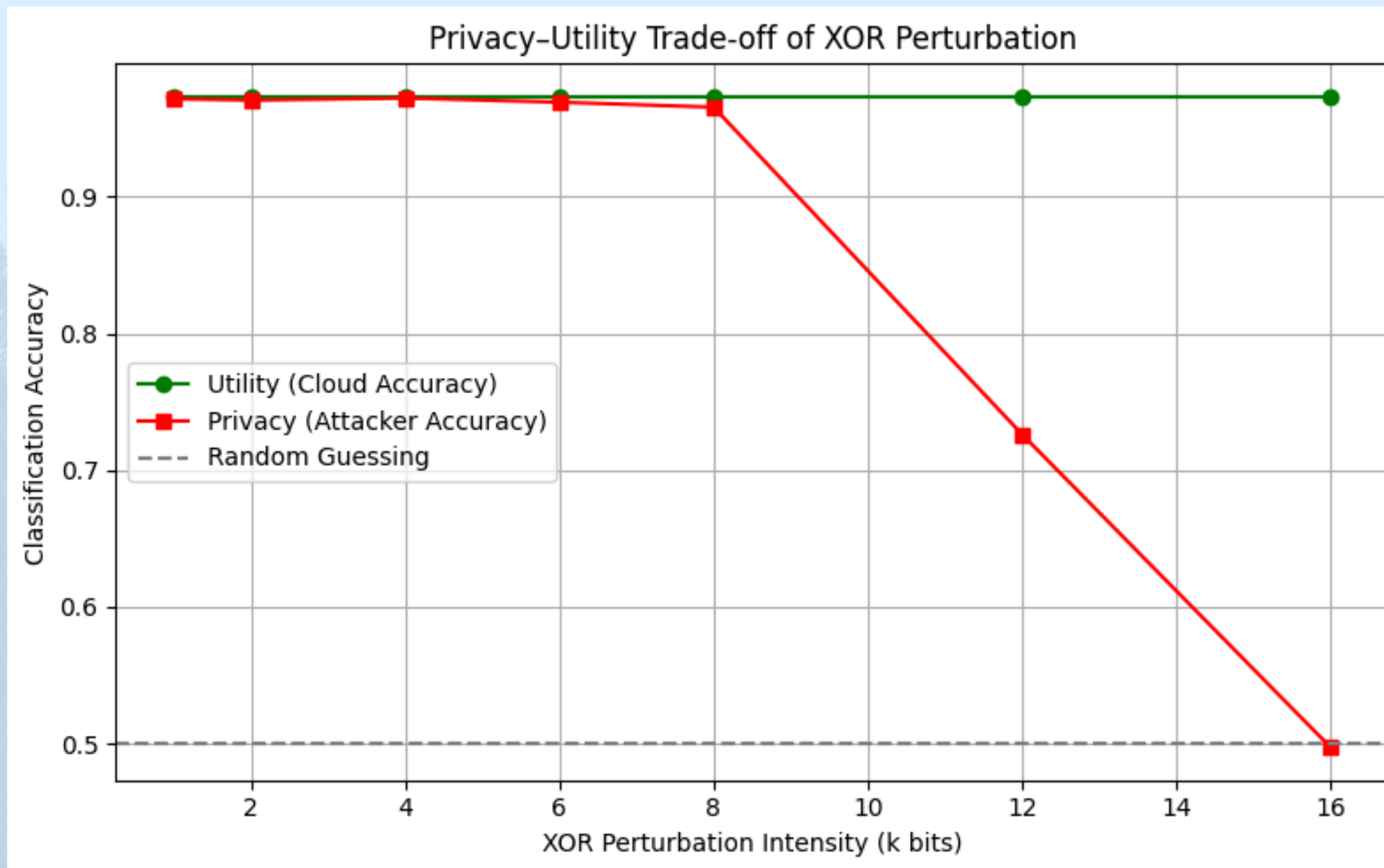


XOR OVER NUMERICAL PERTURBATION?



Numerical vs Bitwise Perturbation bar chart

PARAMETRIC PRIVACY-UTILITY ANALYSIS OF XOR PERTURBATION



THE TRADE-OFF PLOT (K VS ACCURACY)

1. Parameter: XOR Key Bit-Depth K

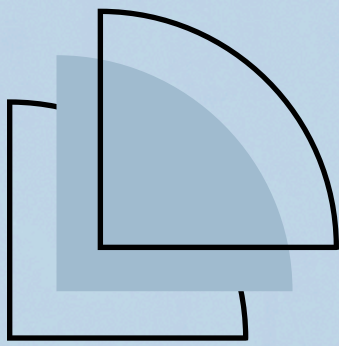
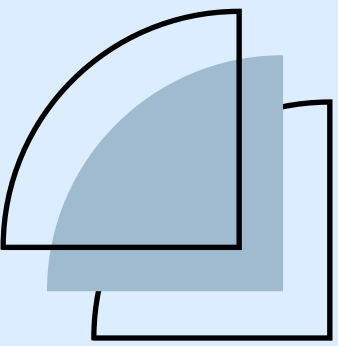
- Controls the "intensity" of the scrambling.

2. Privacy (Red Line):

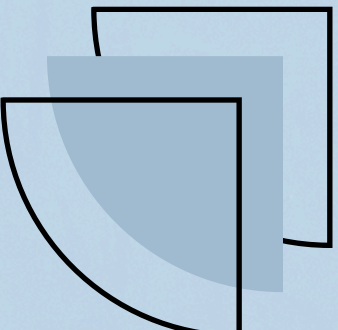
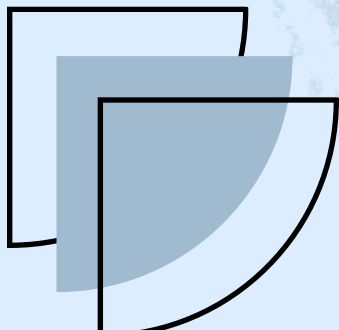
- At low intensity **k=1**, patterns remain visible (High Attacker Accuracy).
- At high intensity **k=16**, patterns are destroyed (Random Guessing).
- Higher k → stronger privacy

3. Utility (Green Line):

- Remains 100% stable regardless of intensity due to perfect reversibility.



CONCLUSIONS

- A lightweight XOR-based reversible perturbation was proposed
 - Operates directly on 16-bit Modbus telemetry
 - Attacker accuracy reduced to ~50% (random guessing)
 - Cloud accuracy preserved at 98.71%
 - No distortion, rounding error, or heavy computation
 - Suitable for legacy and real-time IIoT systems
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The image features four decorative corner elements, each consisting of a light blue square with a darker blue quarter-circle cutout in the corner, positioned at the top-left, top-right, bottom-left, and bottom-right corners. A large, faint, light blue brushstroke graphic runs diagonally across the background from the bottom-left towards the top-right. The text "Thank You" is centered within a large, black-outlined rectangular frame.

Thank You